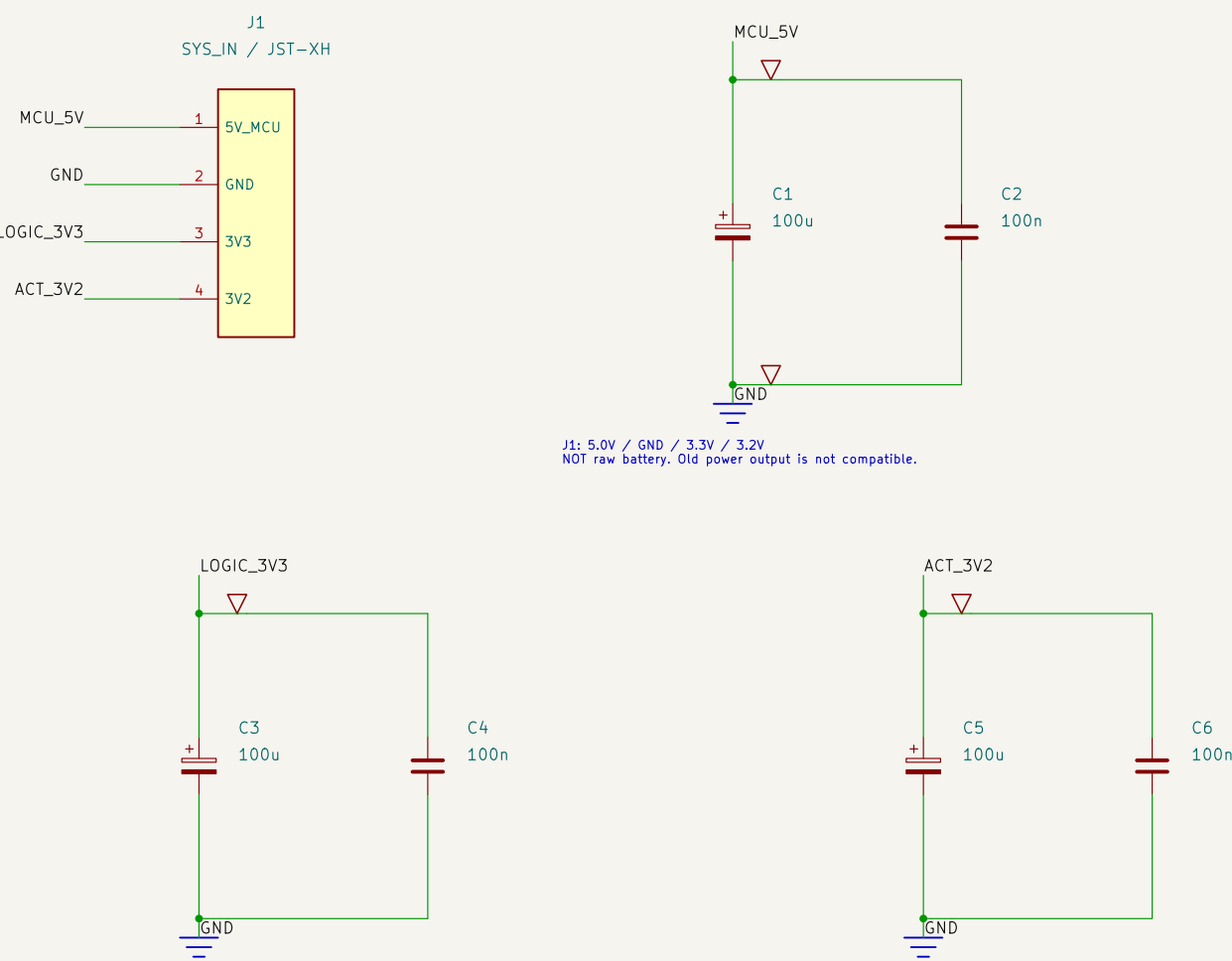


KEYCHAIN KREATURES / MAIN BOARD

ONE-SHEET CIRCUIT VIEW | C.3: RGB mood lamp + top-entry connectors | Engineering prototype

POWER INPUT + LOCAL BYPASS

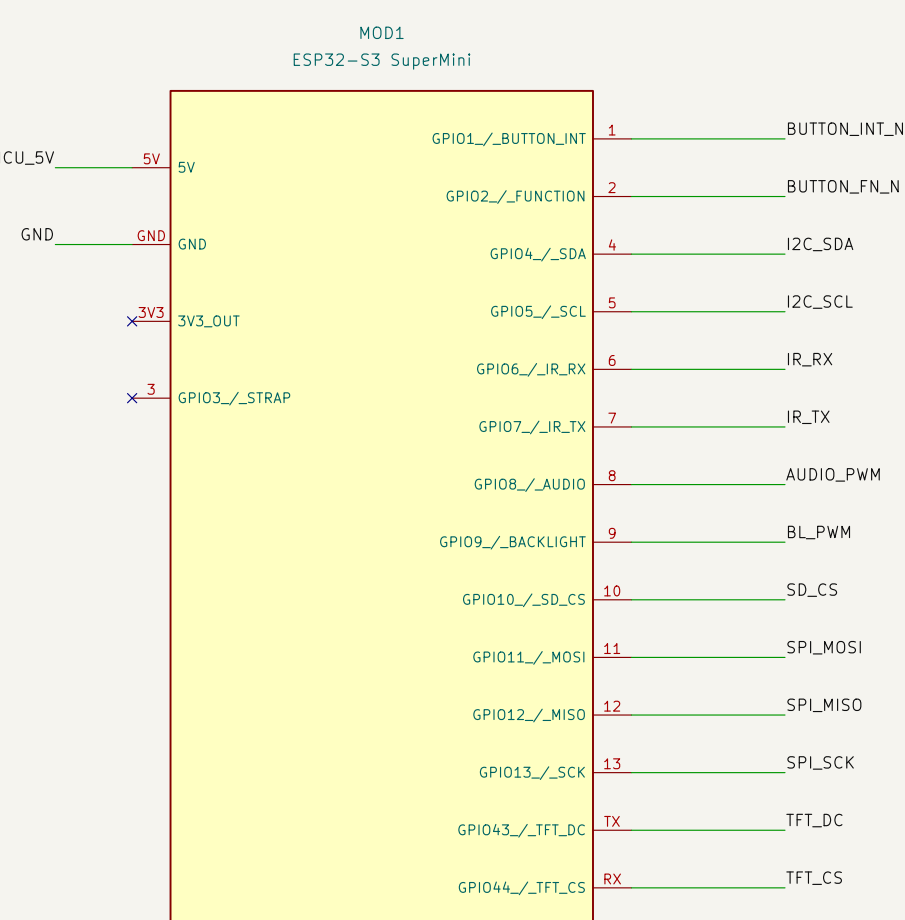
Regulation / charging stay on the separate power board.



J2: 5.0V / GND / 3.3V / 3.2V
NOT raw battery, 5V0 power output is not compatible.

ESP32-S3 SUPERMINI / SOCKET

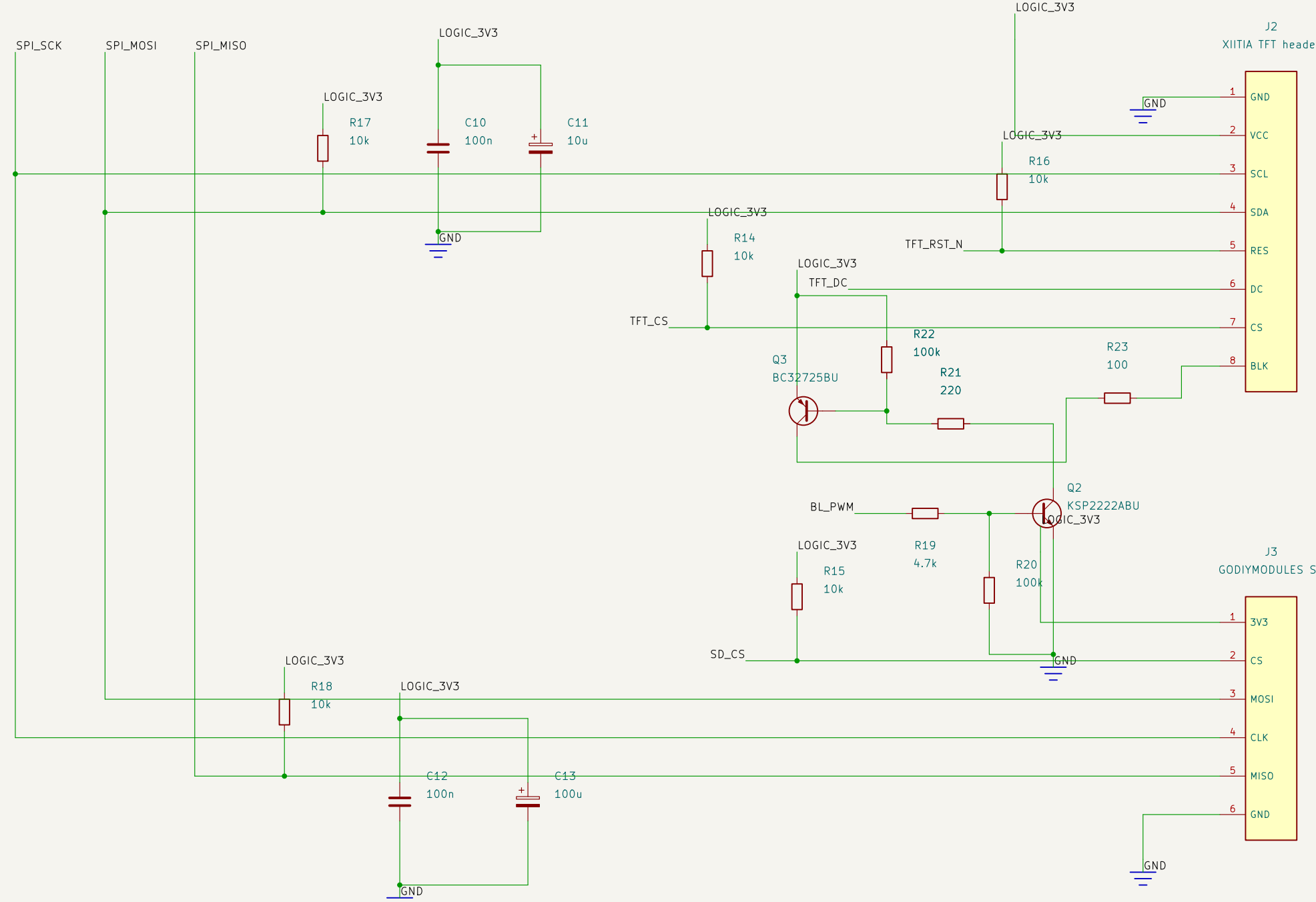
Header signals shown by GPIO; pad names match supplied footprint.



USB service: remove isolated MCU first.
3V3 OUT is intentionally not tied to the external 3V3 rail.

SCREEN + microSD + BACKLIGHT

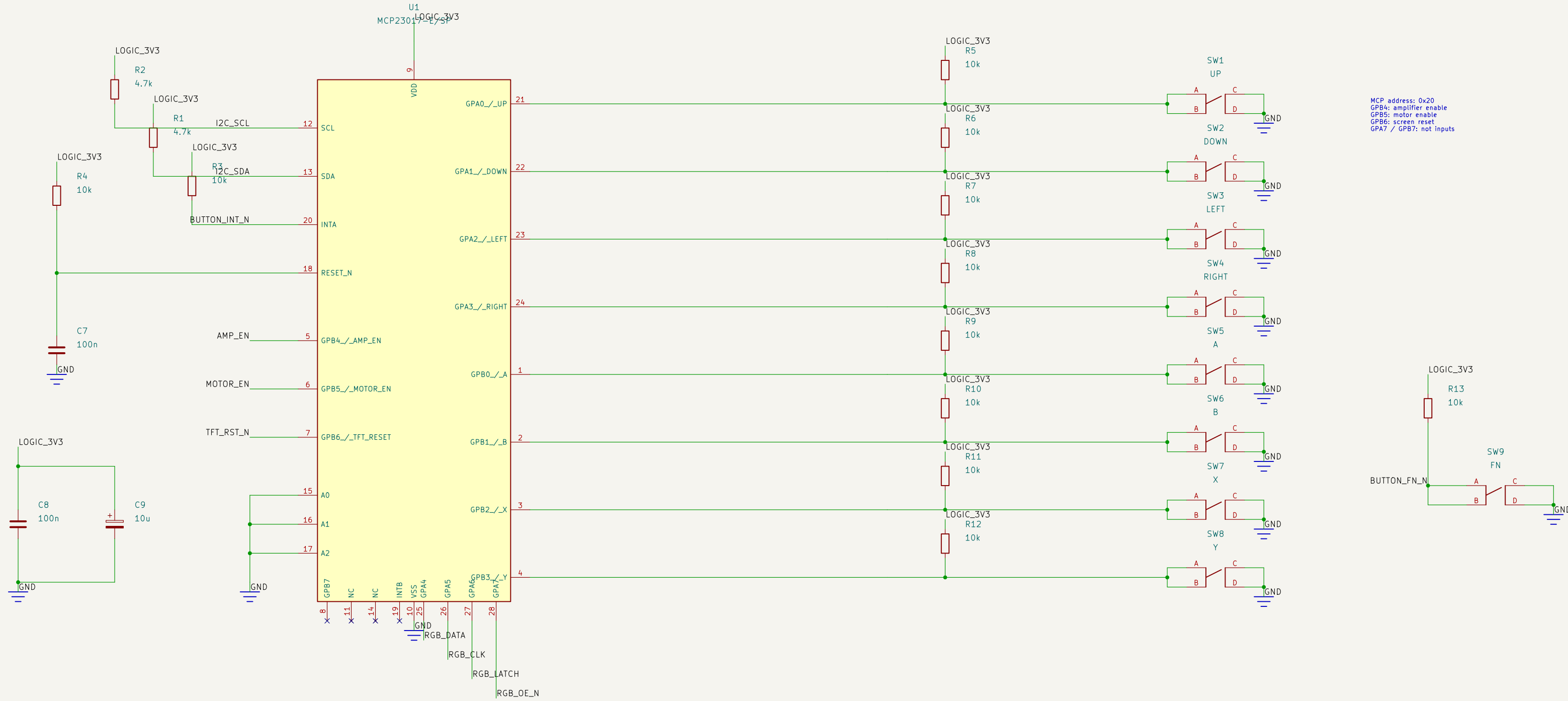
Shared SPI clock / MOSI. Separate CS lines. Both modules use 3.3V logic.



R23 = conservative bridge-up current limit.
Confirm BLK circuitry before raising brightness.

BUTTONS + GPIO EXPANDER

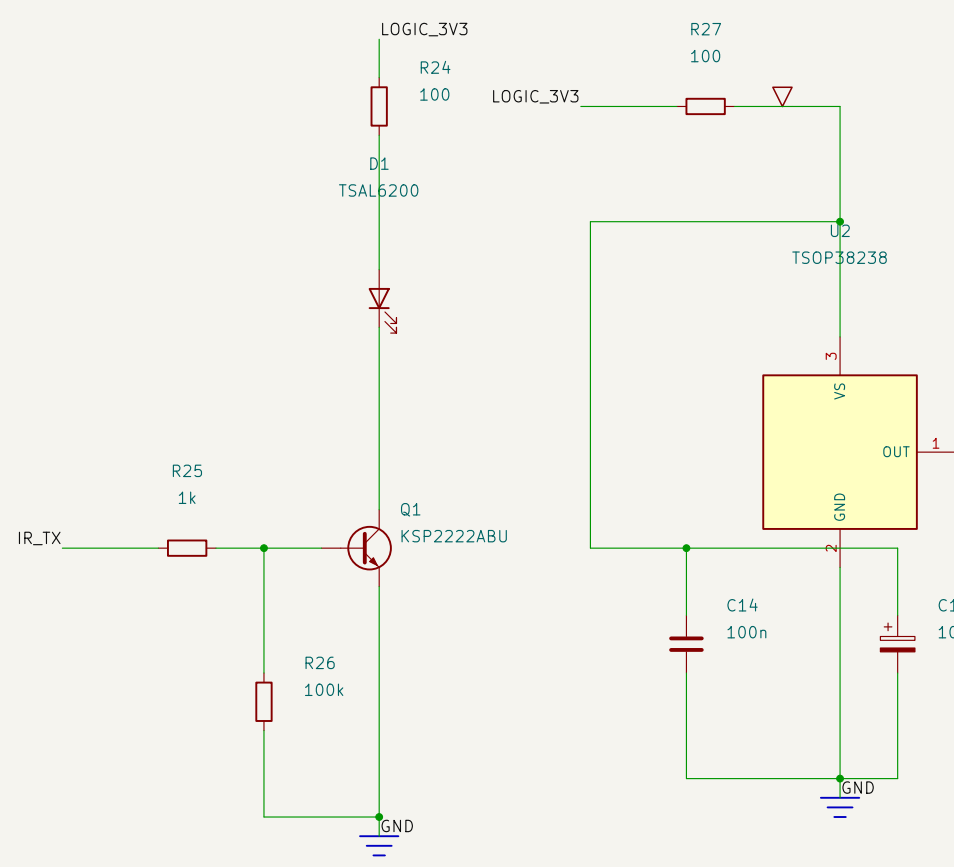
Each switch is wired to its pull-up and input. A/B are one contact; C/D are the other.



MCP address: 0x20
GPIO: amplifier enable
GPIO: motor enable
GPIO: screen reset
GPIO: not inputs

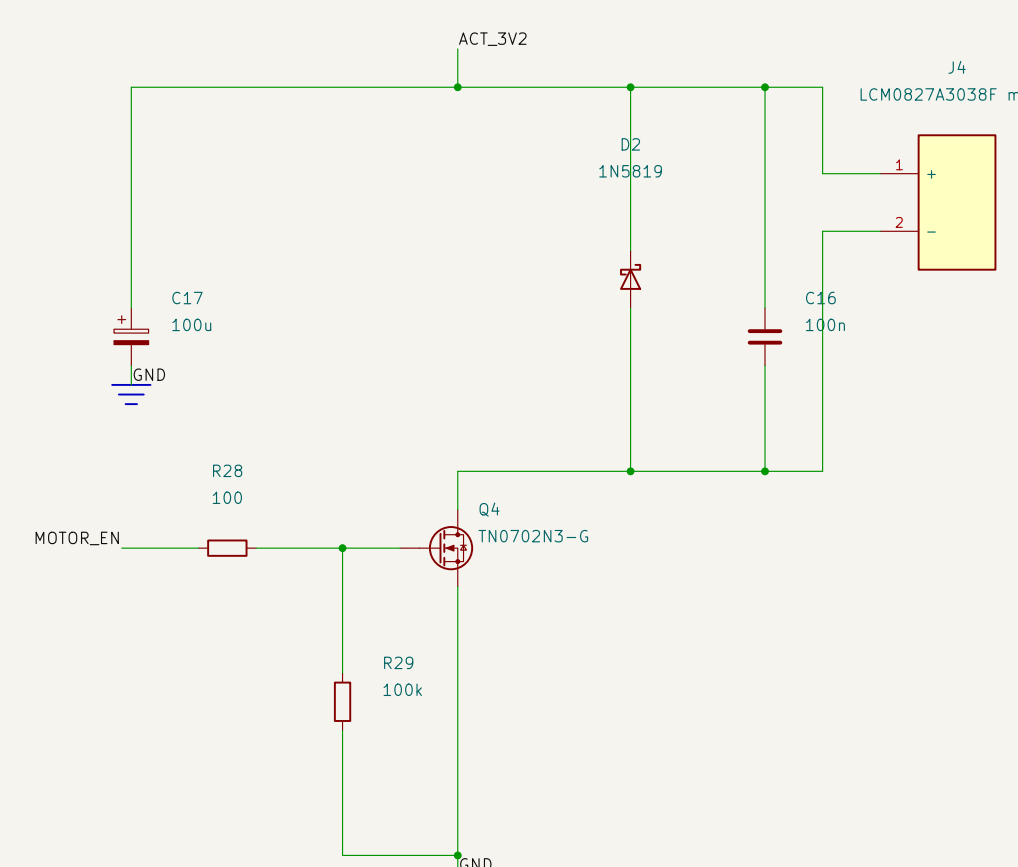
IR TRANSMIT + RECEIVE

38kHz remote-control IR, not sub-GHz RF.



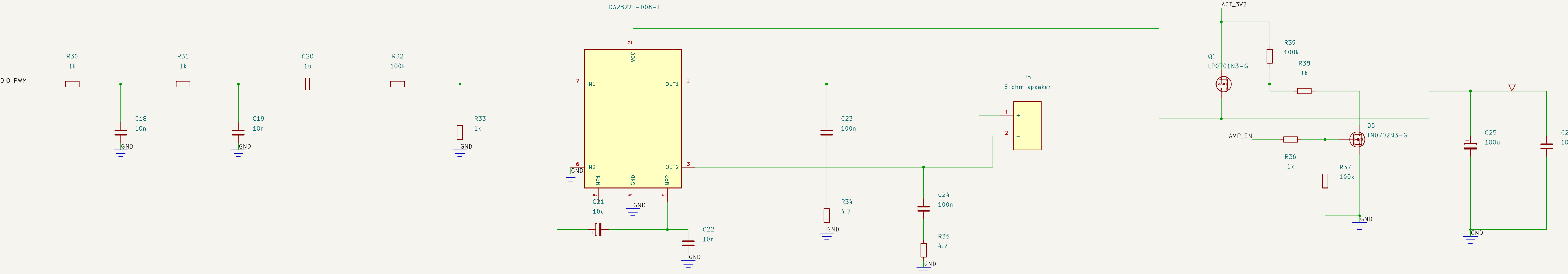
VIBRATION MOTOR

Flyback diode + low-side MOSFET.



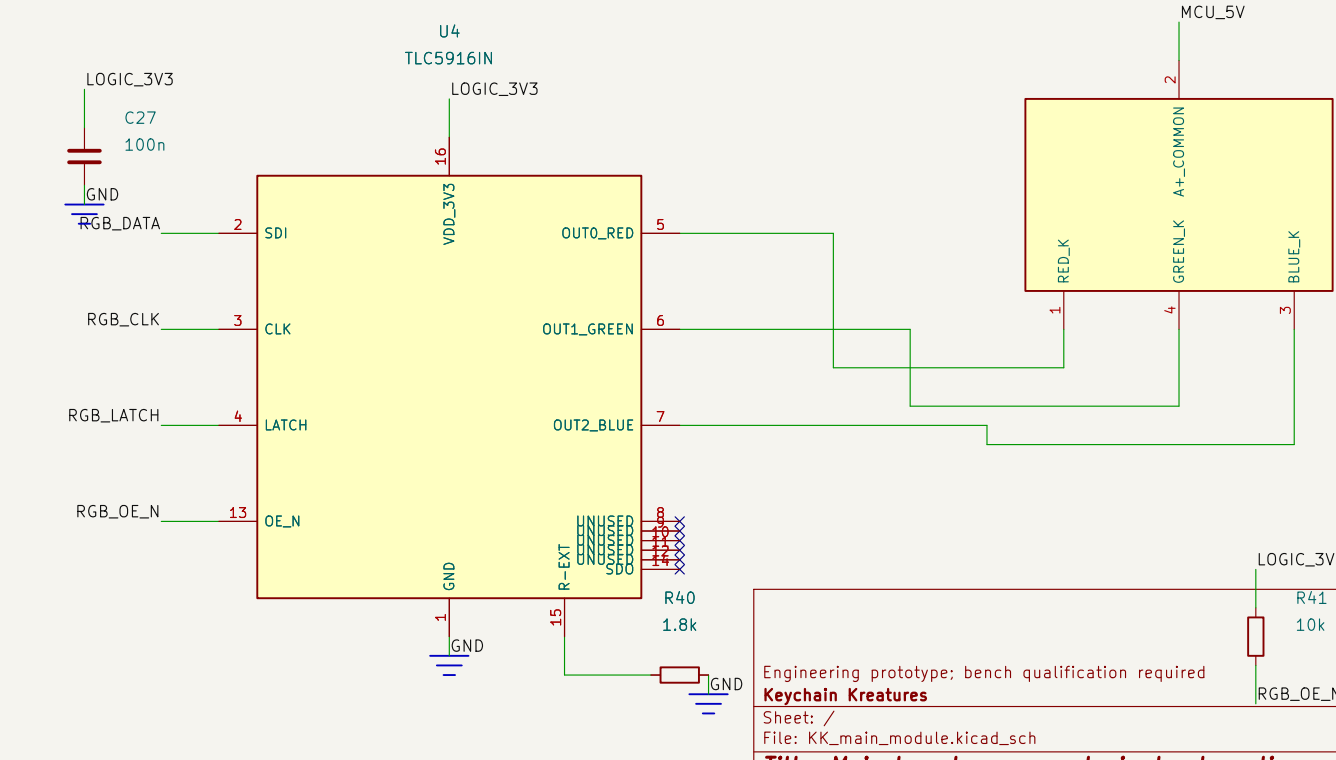
AUDIO / PWM FILTER -> BRIDGE AMPLIFIER -> SPEAKER

Speaker floats between OUT1 and OUT2. Never ground either speaker lead.



RGB MOOD LED / CONSTANT CURRENT

D1 remains IR. U4 drives visible D3; no ESP32 strap pin used.



Engineering prototype: bench qualification required
Keychain Kreatures
Sheet: /
File: KK_main_module.kicad_sch
Title: Main board - grouped wired schematic
Size: A1
Date: 2026-09-10
Rev: C.3 / RGB debug pads.